Step 2: Residue Non-Exhaustion

2.1 Twin Prime Residues Modulo m

For any modulus m, the residue pairs $(n \bmod m, (n+2) \bmod m)$ satisfy:

$n \equiv r_1 \bmod m, \quad (n+2) \equiv r_2 \bmod m, \quad \text{where } r_2 = r_1 + 2 \bmod m.$

For twin primes, residues must avoid divisibility by m, leaving:

- $m - 2$ valid residue pairs.

Example for $m = 5$:

- Valid residue pairs are $(1,3), (2,4), (4,1)$, avoiding $(0,2)$.

By the Chinese Remainder Theorem:

1. Residue pairs for twin primes modulo m persist across all finite sets of primes $\{p_1, \ldots, p_k\}$.
2. These residues cannot be sieved out, leaving infinitely many candidates for twin primes.